Build: Nested Circles (Recursive Component)

Problem: Given a number input, render that many nested circles using recursion.

What This Tests: Recursion in React, dynamic styling, component composition.

jsx

JSX Copy

```jsx
function Circle({ depth, maxDepth }) {
  if (depth > maxDepth) return null;

  const size = (maxDepth - depth + 1) * 80;
  const colors = ['#FF6B6B', '#4ECDC4', '#45B7D1', '#96E6A1', '#DDA0DD', '

  return (
    <div
      style={{
        width: size,
        height: size,
        borderRadius: '50%',
        border: `2px solid ${colors[depth % colors.length]}`,
        display: 'flex',
        alignItems: 'center',
        justifyContent: 'center',
      }}
    >
      <Circle depth={depth + 1} maxDepth={maxDepth} />
    </div>
  );
}
```

```
  return (
    <div style={{ display: 'flex', flexDirection: 'column', alignItems: 'c
      <input
        type="number"
        min={1}
        max={10}
        value={count}
        onChange={e => setCount(Number(e.target.value))}
      />
      <div style={{ marginTop: 20 }}>
        <Circle depth={1} maxDepth={count} />
      </div>
    </div>
  );
}
```

Key Concepts: Recursive component rendering, base case (depth > maxDepth returns null), dynamic sizing and coloring, controlled number input with min/max.

QUICK CHECK — no marks, just for you.

1. In the NestedCircles component, how many circles are rendered before the user touches the number input?

- ◯ 3
- ◯ 1
- ◯ 0
- ◯ 10

Check answer

2. th={1} maxDepth={4} />. Using the lesson's sizing rule, what width does the OUTERMOST circle get?

- 320
- 80
- 400
- 160

Check answer

3. The colors array holds 6 hex values. If maxDepth is large enough to reach depth 6, which border colour does that circle receive?

- #DDA0DD, the fifth colour, since depth is zero-based
- #FFD93D, the sixth and last colour in the array
- #FF6B6B, because the index wraps back to 0
- undefined, because the array only has indices 0 through 5

Check answer

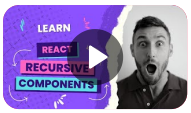
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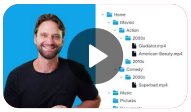
ARTICLES & DOCS

VIDEOS



Recursive component — how & why

YouTube



Build a recursive React component

YouTube



Recursive components in React

YouTube

Hand-picked references for this lesson.